(a) Markov vs. Non-Markov
Markov
Current state is sufficient for Action
Tic-Tac-Toe
Go
Chess
1. Mainly evaluate model reasoning ability
2. Require only current-state understanding
Non-Markov
Reconstruct state from History first, then Act
Match Failed
Filp Back
aC
Match Failed
Filp Back
Facing the same Current State
A B C
a
b
History Info is different
So Action differs
Action: aB
Action: aA
1. Evaluate memory for action ability
2. Require reconstructing hidden state from history
3. Faulty recall leads to wrong actions
(b) RNG Test Suite
Matching Pairs
State N
State N+1
State N+2
State M
State M+1
State M+2
Flip two cards each turn and remove matched pairs
3D Maze
Walking Without Map
Walking With a Minimap
Navigate an egocentric 3D maze from local views
Scale
Performance drops quickly as belief state size grows
Modality
Text works best; 2D/3D visual tracking is harder
Pattern
State tracking is sensitive to visual patterns
MemGap
External memory helps both, more strongly for cards